

Design a brain computer interface gaming system using steady-state visual evoked potential

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Abstract—The brain computer interface (BCI) is an emerging technology for paralyzed people to communicate with external environments through their minds. However, current EEG equipment utilizes metal electrodes, such as gold cup, AgCl, etc., with electrolytes to attach on human skin. These wet-type electrodes are not convenient for use. It usually requires a lousy preparation process and the refill of electrolyte after using one or two hours. In this study, the dry-electrode IMEC headset V1 system was adopted to implement a brain computer interface. Two visual stimuli, flickering at 15 and 17Hz were presented on a LCD screen. Subject's 15 Hz or 17 Hz steady-state visual evoked potential (SSVEP) was induced to control a virtual claw machine. Users gazed at the 15 Hz stimulus to control the movement of the claw and the 17 Hz stimulus was used to clamp objects. During the operation, users gazed at one intended targets to activate its corresponding function. The proposed system has been tested in three volunteers, and the mean accuracy and mean information transfer rate (ITR) were 96% and 18.23 bits/min, respectively. According to our experience, the operation accuracy of the BCI system is related to subject's attention level. Higher concentration level will result in better accuracy. The proposed BCI system can also be used for paralyzed patients to control home facilities for self-caring purpose in the future.

I. INTRODUCTION

Owing to the rapid growth of aging population and rise of social welfare consciousness, the medication care of disadvantage group has drawn great notice. Scientist, physicians and engineers are working together in order to develop assistive technologies for helping paralyzed people. One technology, called brain computer interface (BCI) which measures electrophysiological signals in human brain, is developed for people to communicate or to control environmental devices through their thoughts. The BCI technology is especially useful for handicapped patients, e.g., amyotrophic lateral sclerosis (ALS), severe cerebral palsy, muscular dystrophies, etc., to control external devices in a natural way [1].

Current BCI systems can be divided into two types, one is endogenous BCI and the other one is exogenous BCI. Endogenous BCIs measure movement-related brain signals as control signals. Sensorimotor Mu rhythm, slow cortical potentials (SCPs) and movement-related potential (MRP) were usually adopted in performing imagery movement tasks [2-4]. The Mu rhythm locates within 8Hz~30Hz, while SCP and MRP exist around 1~4Hz. For the exogenous BCI, brain signals were induced by external stimuli to produce specific EEG patterns, such as odd-ball paradigm induced P300 and visually-induced evoked potential (VEP). Both the two exogenous BCIs have drawn great attention, because of their

advantages of less or no training, fast response, easy to implement, high reliability and high accuracy [6-7]. This paper inherits our preliminary results in steady-state visual evoked potential (SSVEP). The proposed system adopted IMEC headset V1 system with dry EEG electrode. Experimental results showed that the IMEC headset V1 system achieved high accuracy in the SSVEP-based BCI implementation. The pin-type dry electrode design achieved excellent signal-to-noise ratio and relative immunity to artifacts. The proposed system has been tested in three volunteers with satisfactory accuracies and information transfer rates (ITR).

II. MATERIALS AND METHODS

A. System Structure

The proposed SSVEP-based BCI system is shown in Fig. 1, in which contains a liquid crystal displayer (LCD), an IMEC EEG headset V1 system, and a desktop PC for signal processing. The LCD presented a virtual claw machine with two flashing flickers, one on the top of the claw (flashing at 15Hz) and the other on the right bottom corner (flashing at 17Hz). The 15Hz flicker was used to move the claw and the 17Hz flicker was used to drop the claw and clamp object. Both the two flickers were designed with 50/50% on-off duties. The user interface (virtual claw machine) was written by LabView software (National Instruments Co., USA) in computer I to achieve real-time feedback and system control. The Oz EEG signals were acquired and processed in computer II using Matlab software (MathWorks). Subject's gazed targets were detected by computer II and the command was sent via TTL signal to activate corresponding function in computer I.

B. IMEC EEG headset and EEG measurement

The IMEC headset V1 system was designed using novel low-power technology. The system equipped with 4 dry EEG electrodes and can be expanded to 8 EEG channels at most. The IMEC system measures high quality EEG signals with high common-mode rejection ratio(CMRR>90dB), low noise (<6 μ v p-p, 0.5-100Hz), and configurable digital filters. The system has the features of lightness and comfortableness, which is suitable for long-time wearing. It utilizes ultra-low-power ASIC technology developed by the technology program (TP) 'analog interfaces' (Yazicioglu et al, ISSCC 2008) with power consumption lower than 5mW. The input impedance of the headset is higher than 80MOhm. One electrode placed at Oz position, referring to International 10-20 system, was

applied with a reference electrode placed at right-ear mastoid and a ground electrode placed at left-ear mastoid.

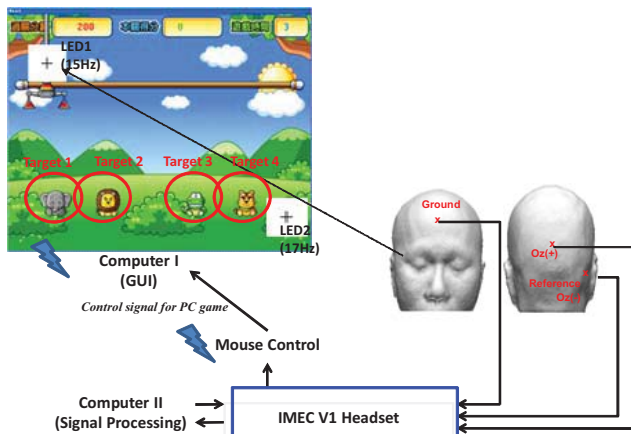


Fig. 1. The schematic diagram of the SSVEP-based BCI gaming system.

C. Subjects and Tasks

In this study, three male participants were recruited. All subjects had no history of clinical visual disease, and their vision were corrected to Snellen visual acuity of 6/6 or better. All subjects signed informed consent form, and this study was approved by the Ethics Committee of Institutional Review Board (IRB), Taipei Veterans General Hospital, Taiwan. EEG data were acquired at 1024Hz sampling rate. Participants were instructed to pick up an object once at one time, until all objects were moved into the basket.

D. Data Analysis

EEG data were segmented into epochs with two-second long. Digitized signals were filtered within 10~20 Hz, and a 2048 points fast Fourier transform (FFT) was performed, which resulted in frequency resolution of $1024/2048 = 0.5$ Hz. The 15Hz and 17Hz peaks on Fourier spectrum were detected and compared. Two thresholds for the 15Hz and 17Hz peaks were determined empirically. Only those epochs with 15Hz and 17Hz spectral peaks exceeded the empirically set threshold were determined and recognized as valid detections.

III. RESULTS AND DISCUSSION

All the three subjects were capable of using the proposed SSVEP-based gaming BCI system with high accuracies. However, according to the frequency preference effect of SSVEP which the amplitude of SSVEP can be varied with different visual stimulation frequency, the threshold for gazed target detection should be determined customized for each participant. The averaged accuracy and information transfer rate (ITR) over the three subjects were 96% and 18.23 bits/min, respectively.

IV. CONCLUSIONS

In our study, we have demonstrated a reliable SSVEP-based

BCI gaming system. The IMEC EEG headset V1 system was used to implement the BCI with high accuracy and ITR. All the three participants had demonstrated their capability to control this two-selection claw gaming system. Compared to other BCI systems which adopt wet-type EEG electrodes, the IMEC dry electrode system is more comfortable and more convenient for users, since no EEG preparation is needed. Besides, some literatures have reported that the SSVEP amplitude can be enhanced in relation to user's attention level. Therefore, the proposed BCI gaming system can be a reliable channel to design brain-controlled consumer electronics as well as a possible way to train user's attention capability.

ACKNOWLEDGMENT

This study was funded by the National Central University and IMEC collaboration project, the Center for Dynamical Biomarkers and Translational Medicine (NSC 102-2911-I-008-001).

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